

Dyness PowerBrick PRO Balancer and Power-Safe BMS Guardian

Engineering implementation, algorithm, commissioning, and recovery reference

Current deployed architecture: telemetry worker, Node-RED controller, and persistent guardian DeviceInstance 101.

Item	Value
Hardware profile	Dyness PowerBrick PRO DIP 00110
Serial protocol	Dyness/Pylon ASCII, 115200 8N1, read-only
Safety fallback	54.0 V CVL / 20.0 A CCL / 100.0 A DCL
Document status	Implementation reference dated 2026-08-19

Contents

Contents	2
1. Purpose and system boundary	3
2. Hardware and communication setup	3
3. RS485 protocol and validation	4
4. Cell 16 reconstruction	5
5. Per-battery capacity SOC and coulomb interpolation	5
6. Balancing controller	6
7. Feed-forward plus slow PI current control	7
8. Centered dynamic-float estimator	8
9. Limit arbitration and permissions	8
10. Power-safe BMS guardian	9
11. Persistence, dashboards, and CSV	9
12. Power-safe deployment and rollback	10
13. Reboot behavior	10
14. Parameter reference	11
15. Acceptance checks	11
16. Implementation references	12

Implementation date: 2026-08-19

This document describes the deployed Cerbo GX solution in the `codex/rs485-dyness-balancer` branch. It distinguishes measured Dyness telemetry from project-specific control policy. It is not a Dyness manufacturer specification and does not replace battery, inverter, or electrical safety documentation.

1. Purpose and system boundary

The installation has three parallel Dyness PowerBrick PRO batteries, addressed 2, 3, and 4. The solution provides:

- Read-only Dyness/Pylon-compatible RS485 telemetry.
- Per-battery cells, current, voltage, status, capacity SOC, and diagnostics.
- Reconstruction of the physical sixteenth cell when only cells 1-15 are sent.
- Automatic selection and current control for passive balancing.
- A persistent Victron D-Bus guardian that remains selected during worker

restarts and RS485 communication failures.

- A fixed communication fallback of 54.0 V CVL, 20.0 A CCL, and 100.0 A DCL.
- Node-RED management pages, event history, persistence, and CSV logging.

The RS485 link is read-only. No Dyness write, wake, configuration, MOSFET, charge, or discharge command is transmitted. Dyness physical protection is the final hardware safety layer.

1.1 Component architecture

Component	Runtime identity	Responsibility
Dyness batteries	RS485 addresses 2, 3, 4	Physical cells, BMS protection, telemetry, and limits
RS485 worker	<code>cerbo-balancer-rs485</code>	Polling, validation, estimation, snapshot, logging
Node-RED controller	<code>node-red-venus</code>	Balancing state machine and requested CVL/CCL
BMS guardian	<code>cerbo-balancer-guardian</code>	Persistent D-Bus publication and communication fallback
Guardian D-Bus service	<code>com.victronenergy.battery.rs485_dyness_guardian</code>	Selected battery/BMS, DeviceInstance 101
Cerbo DVCC	Victron system service	Distributes selected BMS limits to chargers and inverter

Normal data flow is:

```
Dyness RS485 -> telemetry worker -> atomic JSON snapshot -> Node-RED request
-> worker arbitration data -> guardian instance 101 -> DVCC -> chargers
```

The worker runs in telemetry-only publisher mode. The former worker D-Bus instance 100 is not selected and is retained only as a rollback architecture. The guardian at DeviceInstance 101 is the permanent selected battery monitor and controlling BMS.

2. Hardware and communication setup

2.1 PowerBrick PRO DIP switch

Set the five PowerBrick PRO communication DIP switches to:

00110

Read the positions in the order printed on the battery. This setting selects the **Victron and others** protocol profile at **115200 baud**. The deployed serial format is 115200 baud, 8 data bits, no parity, 1 stop bit (115200 8N1).

Confirm switch numbering and ON/OFF orientation against the label on the specific battery revision before energizing the communication link.

2.2 Physical serial interface

The worker uses the stable FTDI path:

```
/dev/serial/by-id/usb-FTDI_FT232R_USB_UART_A602K5MM-if00-port0
```

The runt wrapper invokes the Venus `stop-tty.sh ttyUSB0` handoff before opening the port. A kernel `flock` prevents duplicate workers. The process runs as `nodored:nodored`, while the root wrapper monitors port ownership and reapplies the handoff only if another process opens the adapter.

3. RS485 protocol and validation

The protocol uses printable ASCII frames beginning with `~` and ending with a four-hex-digit checksum plus carriage return. Requests use protocol version `0x20`, CID1 `0x46`, and only read CID2 values.

For an ASCII body with bytes `b_j`, the frame checksum is:

```
checksum = (-sum(j=1..N, b_j)) mod 65536
```

The INFO length is a 12-bit value with a leading nibble checksum:

```
L = 0xABC
length_check = -(A + B + C) mod 16
encoded_length = hex(length_check) || hex12(L)
```

Every response is rejected unless framing, ASCII encoding, checksum, address, CID1, return code, length checksum, INFO length, and CID-specific structure are valid.

3.1 CID2 responsibilities

CID2	Source and content	Use in this implementation
0x42	Addressed battery cells, temperatures, signed current, pack voltage, optional capacity tail	Per-battery telemetry, Cell 16 estimator, capacity SOC
0x44	Status1-Status5, MOSFET states, protection, alarms, cell-check bits	Diagnostics and local eligibility; never an SOC source
0x61	Master system voltage, current, integer SOC, global cell extrema and packed IDs, health and temperatures	System SOC, global extrema bounds, guardian and DVCC authority
0x63	CVL, DVL, CCL, DCL, permission/state byte	Final BMS limits and charge/discharge permissions

Only master address 2 is polled for 0x61 and 0x63. Expected no-reply from slave addresses is not a communication fault. The capacity-derived per-battery SOC comes from the extended tail of 0x42; 0x44 is status-only.

3.2 Units and signed values

```
cell voltage      = raw_u16 / 1000 V
pack voltage      = raw_u16 / 1000 V
battery current   = raw_s16 / 100 A
temperature       = raw_u16 / 10 - 273.15 deg C
capacity          = raw_u24 / 1000 Ah
capacity SOC      = 100 * remaining_mAh / total_mAh
```

Physical validation rejects cells outside 2.0-4.5 V, temperatures outside -40 to 100 deg C, impossible capacity values, malformed structural markers, and incomplete active-battery snapshots.

3.3 Polling and inventory

- Primary telemetry cadence: approximately 8 seconds.
- CID2 0x44 diagnostic cadence: 5 seconds with a bounded timeout.
- Normal discovery: addresses 2-16, completed every 60 seconds.
- Recovery discovery: complete scan every 10 seconds.
- Confirmed removal: 10 consecutive failed complete scans.
- Serial reconnect backoff: 2 seconds.
- Controller and guardian source freshness: 20 seconds.

Normal discovery performs only two short probes per primary cycle. This keeps discovery from monopolizing the serial bus. A partial current sum is never published as authoritative.

4. Cell 16 reconstruction

Cells 1-15 are reported directly at approximately 1 mV resolution. Cell 16 is physical but may not be sent directly. The same battery's pack voltage has approximately 10 mV resolution, so direct subtraction is too noisy for balancing control.

For battery b , let $V_{b,i}$ be directly reported cells 1-15 and $V_{b,pack}$ be that battery's own CID2 0x42 pack voltage.

4.1 Robust common-cell voltage

Sort the 15 direct cells, discard the lowest three and highest three, then average the central nine:

```
ordered = sort(V_b,1 ... V_b,15)
V_b,common = (1/9) * sum(i=4..12, ordered_i)
```

This follows common pack movement while rejecting balancing outliers.

4.2 Raw observation and relative offset

```
V_b,16,raw = V_b,pack - sum(i=1..15, V_b,i)
o_b,raw = V_b,16,raw - V_b,common
```

The raw subtraction is retained for diagnostics only. It never directly drives V_{min} , V_{max} , spread, imbalance detection, or balancing.

Each battery has independent estimator state. No offset, baseline, rolling window, or previous result is shared across addresses.

4.3 Quantization-aware filtered offset

The first valid raw offset becomes the subtraction baseline. Later changes are estimated relative to that baseline. Maintain the five latest raw offsets and use their median as the target:

```
o_b,target = median(last_5(o_b,raw))
e_b = o_b,target - o_b,filtered
```

The combined uncertainty of pack quantization and summing fifteen 1 mV cells is treated as an interval. Errors within approximately 12.5 mV do not force an instantaneous correction. Outside the interval, correction is limited and filtered:

```
e_b,limited = clamp(e_b, -0.020, +0.020) V
o_b,filtered[k] = o_b,filtered[k-1] + g * e_b,limited
```

Normal gain is $g = 0.15$. If the qualifying error remains above 10 mV with the same sign for five consecutive valid samples, gain becomes $g = 0.30$. Acceleration clears immediately when magnitude or sign no longer qualifies.

The unconstrained estimate is:

```
V_b,16,candidate = V_b,common + o_b,filtered
```

4.4 CID2 0x61 global bounds

Fresh global extrema provide physical bounds without imposing the old instantaneous $V_{pack} \pm 5$ mV clamp:

```
V61,min <= V_b,16,published <= V61,max
```

If a packed 0x61 extremum ID identifies this battery and Cell 16, the exact reported extremum is published for that sample. Otherwise the candidate is only range-clamped. Both battery address and cell number must match; another cell at the global minimum must not force Cell 16 to that minimum.

The published Cell 16 participates normally in per-battery extrema, spread, imbalance detection, balancing selection, dashboard telemetry, and CSV data. Diagnostics retain raw, common, target, filtered offset, gain, residual, constraint source, constraint application, and corroboration state.

Estimator state resets after 20 seconds stale, invalid or incomplete cell/pack data, confirmed removal/address replacement, or a pack-voltage jump above 0.5 V. A duplicate sample returns the prior result and does not advance state.

5. Per-battery capacity SOC and coulomb interpolation

The validated 0x42 extended tail contains 24-bit remaining and total mAh. Legacy-only capacity tails are rejected.

5.1 SOC used by balancing

For battery b :

```
SOC_b,raw = 100 * R_b,mAh / T_b,mAh
SOC_b,int = floor(SOC_b,raw)
```

SOC_b,int is the only independent per-battery SOC used for balancing selection and completion. The interpolated value is display and CSV telemetry only. Global charge derating, guardian /Soc, DVCC, and dynamic-float reset continue to use master CID2 0x61 SOC.

5.2 Coulomb integration between capacity steps

For unique samples k-1 and k, trapezoidal charge is:

```
Delta_t_h = (t_k - t_(k-1)) / 3600000
I_avg     = (I_(k-1) + I_k) / 2
Delta_Ah  = I_avg * Delta_t_h
```

The display estimator advances as:

```
SOC_hat[k] = SOC_hat[k-1]
            + 100 * g_direction * Delta_Ah / T_b,Ah
```

Initial gains are g_charge = 1.04 and g_discharge = 1.08. Positive current uses charge gain; negative current uses discharge gain.

Whenever remainingCapacityRawMah changes below 100%, the estimator:

1. Evaluates the completed interval for gain learning.
2. Re-anchors SOC_hat to SOC_b,raw.
3. Clears accumulated interval Ah.
4. Resumes integration until the next raw capacity transition.

With the observed 280 Ah packs and 2.8 Ah capacity steps, raw anchors coincide with integer percentage transitions. The code triggers on changed remaining mAh and anchors to raw percentage, not explicitly to the floored integer.

5.3 Directional gain learning

For a valid interval:

```
Delta_Ah_observed = (R_k,mAh - R_anchor,mAh) / 1000
g_observed        = Delta_Ah_observed / Delta_Ah_integrated
g_new             = 0.80 * g_old + 0.20 * g_observed
```

Training requires matching signs, samples no more than 20 seconds apart, below-100% endpoints, and $0.90 \leq g_{\text{observed}} \leq 1.20$. An implausible interval re-anchors without training and emits SOC_CAPACITY_CORRECTION.

5.4 Voltage-based full anchor

Capacity-reported 100% alone does not anchor interpolation. For each battery independently:

```
if max(effective_cells_b) >= 3.500 V:
    SOC_hat = 100.000%
    interval_Ah = 0
```

The effective 16-cell vector includes reconstructed Cell 16. While the condition remains true, positive charging current is not integrated. When Vmax falls below 3.500 V, discharge integration can reduce SOC immediately.

Invalid capacity data or a gap above 20 seconds discards unfinished integration but retains learned gains. State and gains are written atomically with a validated backup.

6. Balancing controller

The Node-RED controller has NORMAL, BALANCING, and SAFETY_STOP states. Automatic balancing defaults ON after a fresh state or Restore Defaults.

6.1 Entry qualification

For each battery independently, entry requires eight consecutive unique valid telemetry samples satisfying all of:

- Per-battery spread strictly greater than 30 mV.
- Charge MOSFET ON.

- No local hard protection.
- Valid complete telemetry and valid independent integer capacity SOC.
- Completion latch not blocking automatic selection.

Exactly 30 mV does not qualify. Failure of any requirement resets only that battery's entry counter. Among qualified batteries, deterministic ascending address order is used.

6.2 Selection and immediate exits

The selected address remains locked during balancing. Full-SOC and protection exits are immediate. If a local charge path is interrupted, only that battery is released; normal charging for the other parallel batteries is preserved.

All expected batteries at SOC_{b, int} = 100% complete and latch the full-SOC session. The latch rearms only after master charge permission is observed OFF and then continuously ON for 5 seconds.

6.3 Qualified low-spread exit

The selected battery exits balancing after eight consecutive unique samples:

```
I_selected > 1.5 A
spread_selected < 0.030 V
```

Exactly 1.5 A or 30 mV does not qualify. Any failed sample resets the exit counter.

6.4 Non-balancing SOC charge-current caps

In NORMAL, master CID2 0x61 SOC selects an editable per-battery CCL ceiling. The controller multiplies it by the confirmed battery inventory:

$$I_{\text{CCL}, \text{SOC}} = I_{\text{per battery}}(\text{SOC}) \cdot N_{\text{confirmed}}$$

SOC	Default per battery	Three-battery aggregate
below 97%	none	none
97%	15 A	45 A
98%	6 A	18 A
99% and above	3 A	9 A

$N_{\text{confirmed}}$ is the count of unique configured expectedAddresses, not the number of batteries responding to the latest poll. This prevents a temporary missing response from lowering the multiplier. Each value is editable on the controller page and validated as greater than zero and at most 100 A/battery.

These caps do not apply in BALANCING, where selected-battery current control is authoritative subject to final BMS, UI, thermal, and permission ceilings.

7. Feed-forward plus slow PI current control

The controller regulates selected-battery current by changing aggregate CCL. It does not assume equal sharing among parallel batteries.

Let I_s be selected-battery current, I_{pos} the sum of positive battery currents, and $I_{\text{target}} = 2.0$ A.

When $I_s \geq 0.25$ A and $I_{\text{pos}} \geq 0.5$ A:

```
share_raw[k] = clamp(I_s / I_pos, 0.001, 1.0)
share_f[k]   = alpha * share_raw[k] + (1-alpha) * share_f[k-1]
alpha       = 0.20
```

Feed-forward and control error are:

```
I_ff           = I_target / share_f
I_ff, effective = K_ff * I_ff
e              = I_target - I_s
P              = Kp * e
```

The discrete integral approximates the continuous integral:

```
I_term(t) = clamp(Integral[t0,t](Ki * e(tau) d tau), -Imax, +Imax)
```

```
I_term[k] = clamp(I_term[k-1] + Ki * e[k] * Delta_t,
                  -Imax, +Imax)
```

Production defaults are $K_{ff} = 1.0$, $K_p = 0.20$, $K_i = 0.02$, and $I_{max} = 10$ A. The unconstrained request is:

```
I_request_raw = I_ff, effective + P + I_term
I_request      = clamp(I_request_raw, aggregate_min, aggregate_max)
```

Upward movement is slew-limited to 10 A/min:

```
I_request[k] <= I_request[k-1] + (10 A/min) * Delta_t_min
```

Downward correction is immediate. Anti-windup prevents positive integral growth while the output is saturated. Duplicate Node-RED ticks return the held request and do not integrate twice.

7.1 Solar-limited pause

After four deficient telemetry samples, the controller pauses PI and feed-forward updates without releasing selection when both selected current and total current demonstrate insufficient available charging power. Recovery also requires four qualifying samples. BMS-limited, thermal-limited, permission-off, non-positive selected current, and CCL-zero states freeze control rather than integrating against an unavailable actuator.

8. Centered dynamic-float estimator

The dynamic float uses master CID2 0x61 system pack voltage and global Vmax. The cell trigger is editable on the controller page as floatCellVoltageThreshold.

```
default trigger = 3.450 V/cell
allowed range   = 3.000 ... 3.650 V/cell
```

Changing the trigger invalidates the old learned float and clears incomplete acquisition state.

8.1 Centered 4+1+4 acquisition

Maintain a four-value ring buffer while Vmax is below the trigger. At the first accepted crossing, form a window with:

```
[four samples before] + [crossing sample] + [four samples after]
```

The learned voltage is the arithmetic mean:

```
V_float = (1/9) * sum(k=-4..+4, V61, pack[k])
```

The four post samples are accepted only while Vmax is at or above the trigger. If Vmax falls below after crossing, acquisition pauses and preserves the partial window; a later qualifying sample resumes it. A crossing without four pre-trigger samples is ignored and must be rearmed below threshold.

Invalid, stale, incomplete, or duplicate telemetry cannot add a sample. Invalid telemetry resets an incomplete window but retains a qualified float. Master SOC at or below 98% resets the learned voltage to the 56.5 V default and clears all transient acquisition state.

8.2 Float application

The learned float becomes a CVL candidate only when full-SOC balancing has completed, the completion latch is set, and master CID2 0x61 SOC is 100%. It is not used as the balancing CVL during active balancing.

Final normal CVL is:

```
CVL_normal = min(56.5 V hard ceiling,
                  valid Dyness CVL,
                  enabled Cerbo UI CVL,
                  learned/default float when active)
```

9. Limit arbitration and permissions

The controller proposes values; the runtime independently applies lower authoritative ceilings.

In normal operation:

```
CCL_candidate = min(controller request,
                     valid Dyness CCL,
                     enabled Cerbo UI CCL,
                     applicable normal SOC cap)
```



```
CCL_effective = CCL_candidate * thermal_factor
```

In balancing, the normal SOC cap is omitted. Valid Dyness permission OFF or CCL zero forces charge allowance to zero. DCL and discharge permission remain independent of charge control. The service never converts missing limits into invented high limits.

Standard D-Bus outputs include:

- /Info/MaxChargeVoltage
- /Info/MaxChargeCurrent
- /Info/MaxDischargeCurrent
- /Bms/AllowToCharge
- /Bms/AllowToDischarge
- /Dc/0/Voltage, /Dc/0/Current, /Dc/0/Power
- /Soc

Diagnostic /Control/* paths expose requested values, effective values, freshness, thermal factor, authority, and reason.

10. Power-safe BMS guardian

The guardian is an independent locked runit process. It reads the worker's atomic latest snapshot once per second and remains the selected D-Bus battery through worker updates, failures, and restarts.

10.1 Source validity

A source sample is valid when:

```
snapshot.valid == true
source age <= 20000 ms
CID2 0x61 SOC is an integer in 0..100
effectiveControl.outputValid == true
```

10.2 Guardian state machine

Mode	Condition	Published behavior
BOOTSTRAP	No persisted or live last-good snapshot	Ready false; do not synthesize SOC
NORMAL	Fresh valid source and recovery complete	Publish worker effective values
FALLBACK	Missing, malformed, invalid, or stale source	Publish fixed 54 V / 20 A / 100 A fallback
RECOVERY	First valid unique sample after fallback	Keep fallback until second valid unique sample

Fallback outputs are:

```
CVL = 54.0 V
CCL = 20.0 A
DCL = 100.0 A
AllowToCharge = true
AllowToDischarge = true
Connected = true
State = RS485_FALLBACK_54V_20A
```

Last valid CID2 0x61 SOC is retained indefinitely and never replaced by a synthetic startup zero. Live VE.Bus/MultiPlus voltage and current are used when available; otherwise last-good measurements are retained. A communication alarm and guardian diagnostics identify fallback mode and source age.

Return to NORMAL requires two consecutive complete, fresh, unique worker samples. The last-good snapshot, SOC, and timestamp are persisted atomically with a validated backup.

11. Persistence, dashboards, and CSV

Important runtime paths are:

```
/data/home/nodered/cerbo-balancer-latest.json
/data/home/nodered/cerbo-balancer-state.json
```

```

/data/home/nodered/cerbo-balancer-capacity-soc.json
/data/home/nodered/cerbo-balancer-guardian-state.json
/data/home/nodered/cerbo-balancer-rs485-inventory.json
/data/home/nodered/cerbo-balancer-events.jsonl
/data/home/nodered/cerbo-balancer-csv/
/data/home/nodered/cerbo-balancer-telemetry/

```

Detailed parsed telemetry is retained for 24 hours. Compact summary JSONL is written every 60 seconds and retained for 30 days. Raw RS485 frames are not persisted.

The maintenance page displays system telemetry, global extrema, every effective cell, Cell 16 estimator diagnostics, per-battery integer and interpolated SOC, capacity anchors, learned gains, status bits, temperatures, inventory, serial ownership, and communication health.

CSV files use a fixed schema and fixed battery inventory for each recording. A new schema starts a new file; columns are not appended to an active legacy schema. The current schema includes per-battery raw/integer/interpolated SOC, anchor state, mAh/Ah, gains, Vmax/full-reset state, cell values, Cell 16 diagnostics, status bytes, controller terms, float state, authority, and final effective limits.

12. Power-safe deployment and rollback

12.1 Worker-only deployment

1. Require guardian DeviceInstance 101 to be selected, Ready, and NORMAL.
2. Back up the worker and current latest snapshot.
3. Copy the new worker to a temporary file and validate Python syntax.
4. Preserve nodered:nodered ownership and atomically rename it into place.
5. Restart only cerbo-balancer-rs485 with `svc -t`.
6. Confirm guardian immediately enters FALLBACK with exactly 54/20/100.
7. Confirm a new worker PID and two valid samples before NORMAL resumes.
8. Roll back only the worker if recovery fails; leave guardian running.

12.2 Node-RED deployment

1. Back up live flows.json and flows_cred.json.
2. Merge generated controller/dashboard nodes by node ID into the complete live flow; never replace the 81-node live flow with a partial export.
3. Validate staged JSON, preserve ownership, and atomically replace flows.
4. Restart only node-red-venus.
5. Allow for the Cerbo's palette-loading startup delay.
6. Confirm dashboard HTTP 200 and unchanged worker/guardian PIDs.
7. Restore the flow backup independently if Node-RED fails.

12.3 Guardian upgrades

Do not restart the active guardian for ordinary worker or Node-RED deployments. Upgrade guardian code only through a validated shadow/A-B instance while the existing selected guardian remains operational. Perform selection migration with external AC or bypass when physical continuity is required.

13. Reboot behavior

`/data/rc.local` recreates runit service directories under volatile storage. It registers the guardian before the worker, creates bounded multilog services, and then links both supervisors into `/service`. Each process has an independent lock and log rotation (25 kB, four files).

After reboot:

1. Guardian restores last-good state and selected instance 101.

2. Guardian publishes fallback until worker telemetry is valid.
3. Worker claims the FTDI adapter and starts telemetry-only polling.
4. Two unique valid snapshots return the guardian to NORMAL.
5. Node-RED restores controller configuration and state independently.

14. Parameter reference

Parameter	Default	Meaning
RS485 serial	115200 8N1	PowerBrick PRO Victron/others profile
Poll interval	8 s	Primary complete telemetry cadence
Freshness	20 s	Controller and guardian validity bound
Balance spread	>30 mV	Strict entry threshold
Qualification	8 samples	Entry and selected low-spread exit
Selected target	2.0 A	Balancing current target
Exit current	>1.5 A	Strict selected-battery exit criterion
Feed-forward alpha	0.20	Current-share EWMA
Kp / Ki	0.20 / 0.02	Slow PI defaults
Upward slew	10 A/min	Aggregate request rise limit
SOC CCL at 97%	15 A/battery	Editable; multiplied by confirmed inventory
SOC CCL at 98%	6 A/battery	Editable; multiplied by confirmed inventory
SOC CCL at >=99%	3 A/battery	Editable; multiplied by confirmed inventory
Float trigger	3.450 V/cell	Editable, allowed 3.000-3.650 V
Float window	4+1+4	Centered pack-voltage samples
Float reset SOC	<=98%	Clear learned float to 56.5 V
Cell 16 median	5 samples	Raw-offset target filter
Cell 16 gain	0.15 / 0.30	Normal / persistent-error correction
Cell 16 correction	+/-20 mV	Per-update error clamp
Capacity gains	1.04 / 1.08	Initial charge / discharge gains
Gain EWMA	0.20	Capacity-interval learning rate
Guardian fallback	54 V / 20 A / 100 A	CVL / CCL / DCL
Guardian recovery	2 samples	Unique complete samples before NORMAL

15. Acceptance checks

- DIP code is 00110; serial operation is 115200 8N1.
- All active addresses return complete valid 0x42; only master answers 0x61.
- Cell 16 uses independent filtered state and respects correctly decoded global

bounds without being pinned to another cell's extremum.

- Per-battery capacity SOC anchors on each remaining-mAh transition below 100%.
- Interpolated SOC never affects guardian SOC, DVCC, global CCL, or float reset.
- Balance entry and low-spread exit change only at sample eight.
- Float qualification changes only at centered sample nine.
- Changing the float trigger resets old learning.

- Worker restart leaves guardian PID and D-Bus service continuously present.
- RS485 disconnection produces persistent 54/20/100 fallback, never CCL zero.
- Recovery requires two valid unique samples.
- Reboot recreates both runit services and returns guardian to NORMAL.
- Dashboard and CSV identify raw, estimated, bounded, requested, and effective

values without conflating their authority.

16. Implementation references

Primary implementation paths:

scripts/dyness_rs485_protocol.py	protocol framing and CID2 decoding
scripts/dyness_rs485_service.py	worker, estimators, arbitration, logging
scripts/dyness_bms_guardian.py	persistent D-Bus guardian
src/controller.js	balancing and dynamic-float controller
src/controller_dashboard.html	controller management interface
src/maintenance_view.html	RS485 maintenance interface
deploy/cerbo-rc.local	guardian-first reboot registration
flow/cerbo-balancer-controller.json	generated Node-RED flow

The implementation and thresholds in this document describe the project as of 2026-08-19. Revalidate the manual whenever control logic, protocol decoding, fallback values, D-Bus selection, or hardware configuration changes.